

Jack Wiegman

Software Developer

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Summary

Computer Science student with 8+ years of self-taught programming experience. Love to build and explore systems. Developed an 8-bit assembler with virtual memory execution, and implemented a PostgreSQL backend with CMake build system. Comfortable at the command line with familiarity in both Unix and Windows environments.

Technical Skills

Languages:

C, C++, Python, Java, Bash, SQL, Lua, Assembly

Tools:

Linux, Git, CMake, GDB, Windows Server, SQL Server, SQLite

Concepts:

Computer Architecture, Compiler/Assembler Design, Data Structures, Algorithms, Database Design, Network Fundamentals

Projects

8-Bit Assembler & Virtual Machine

C, Assembly [GitHub]

- Designed custom 8-bit CPU simulator with custom instruction set and byte-addressable memory
- Developed assembler parsing assembly source files and generating machine code
- Implemented virtual memory address space executing compiled machine code instructions
- Built register file, stack pointer, and full execution cycle

Team Billing System

C++, PostgreSQL, CMake, Git [GitHub]

- Worked collaboratively with a team to design and build a mock billing system using Git Enterprise
- Designed and implemented Database and API classes with PostgreSQL integration using libpq
- Built database connection layer and helper functions for CRUD operations
- Configured CMake build system with cross-platform presets (Linux/Windows) and libpq dependency integration
- Set up PostgreSQL database schema and connection infrastructure using Supabase

Linux Dotfiles

Bash, Lua, JSON [GitHub]

- Comprehensive system configuration for Arch Linux environment
- Configured systemd boot loader/init, disk partitions, and hyprland window manager
- Custom workflow with CLI tool, terminal emulator, and development tool configurations
- Neovim text editor configuration with LSP and debug adapter support, formatting, and auto-complete

Experience

IT Developer & System Administration

University Bicycles, Boulder, CO

January 2023 – Present

- Led Windows 10 to 11 migration across 10+ workstations with minimal operational downtime
- Administer Windows Server 2012 with SQL Server database for retail management software
- Perform database backup operations and integrity validation using SQL query scripts and DBCC commands
- Diagnosed backup issues and disk failure using Linux recovery tools
- Maintain e-commerce website on Workstand platform, implementing minor HTML/CSS modifications
- Provide end-user support and hardware/software troubleshooting for staff and customers
- Managed hardware procurement, deployment, and system upgrades for point-of-sale infrastructure

Education

University of Colorado Boulder

2020 – 2022

- Completed coursework in Computer Science fundamentals

Front Range Community College, Larimer, CO

Associate of Science in Computer Science

20-Credit Programming Certificate

Expected: May 2026

Completed (conferring May 2026)

- Relevant coursework: Computer Architecture, Python, Java, Advanced C++, Object Oriented Programming

Additional

- 8+ years programming experience (self-taught since high school)
- Advanced Mathematics: Calculus III, Differential Equations
- Interest in linux/open-source communities, cybersecurity, and CTF competitions